The Spectral Gap-Statement:

When the Negative Subspace of Attention Transport is a Well-Posed Invariant

Evgeny Vyaltsev Daniil Vyaltsev

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Abstract

The Φ-Transport framework casts attention as a convex transport process whose free-energy Hessian governs an instability of mass transport: a negative eigenvalue produces an exponentially growing mode. The adaptive rank-local router built on that framework implicitly assumes that the *number* of such modes, $\dim E_-$, is a well-defined integer. This paper makes that assumption precise. We show that $\dim E_-$ is a topological invariant — threshold-free via the Riesz spectral projector, and stable under perturbations of operator norm below half the spectral gap, by the Davis–Kahan theorem — exactly when the free-energy Hessian possesses a gap, $\Delta(H) > 0$. The three empirical operating "windows" of the framework (energy, separation, softness) are shown to be three coordinate sections of this single spectral condition. The link from the invariant to geometric defocusing requires, in addition, an energy-window restriction on the Jacobi–Maupertuis curvature, which we state precisely. Finally, we validate the picture on a trained Llama-3.2-1B: the gap-regime holds for 96% of attention heads, it is established by an explosive phase transition at the first layer, and the typical head carries exactly one dominant transport mode. The geometry the framework predicts is not a synthetic artefact; it is the typical state of attention in a trained transformer.

1 Introduction

This paper adopts the Φ-Transport viewpoint: the observation that several machine-learning primitives — empirical sorting, radix sorting, equal-mass occupancy, and LogSumExp attention — can be cast as special cases of a single convex transport potential Φ. A consequence of that viewpoint is an inertial Wasserstein flow whose damped dynamics are governed by a characteristic polynomial $z^2 + \gamma z + \kappa = 0$: when the free-energy Hessian has a negative eigenvalue $\kappa < 0$, the flow has an exponentially growing mode, and transport mass defocuses.

The framework's adaptive rank-local router exploits this: it routes a query to a low-rank sparse path when the effective number of unstable transport directions is small. That number is the dimension $\dim E_-$ of the negative subspace of the free-energy Hessian. The router's empirical law, $r^\star \approx d_{\text{eff}} - 1$, treats $\dim E_-$ as a well-defined integer — but a count of negative eigenvalues is only well-posed if the spectrum actually separates the negative modes from the rest. When it does not, the count depends on an arbitrary tolerance, and the router's premise dissolves.

This paper supplies the missing precision. Its single technical object is the *spectral gap* $\Delta(H)$ of the free-energy Hessian H, and its single thesis is that $\dim E_-$ is a meaningful invariant exactly inside the regime $\Delta(H) > 0$. The contributions are:

- A threshold-free definition of $\dim E_-$ via the Riesz spectral projector, and a proof (Proposition 1) that it is well-posed precisely in the gap-regime.

- A perturbation-stability theorem (Proposition 2): $\dim E_-$ is preserved under any symmetric perturbation of operator norm below $\frac{1}{2}\Delta$, by Davis–Kahan.

- The unification of the framework's three empirical operating windows — energy, separation, softness — as three coordinate sections of the single condition $\Delta(H) > 0$.

- A precise energy-window caveat (Proposition 3) on the implication "negative Hessian ⇒ negative curvature": it is false without an energy restriction, and we give the exact threshold.

- An empirical validation on a trained Llama-3.2-1B (§3.9): the gap-regime is the typical state of attention, established by an explosive phase transition at layer 1.

The body of the paper is Section 3. Section 2 first recaps, self-containedly, the minimal Φ-Transport material the argument depends on.

2 Preliminaries: the free-energy potential and the inertial flow

This section develops, self-containedly, the three elements of the Φ-Transport viewpoint that the Gap-Statement builds on: the free-energy potential, the inertial flow it induces, and the role of damping. The material is elementary — the central identity is the Hessian of the cumulant-generating function of an exponential family — and is given here so that the paper depends on no external source.

The free-energy potential. Φ-Transport represents a family of machine-learning primitives by a single convex potential. For the case of interest here, the potential is the free energy

$$\Phi_\beta(x) \;=\; \frac{1}{\beta}\,\log\sum_j e^{-\beta U_j(x)}, \tag{1}$$

with β an inverse temperature and U_j a family of component potentials. LogSumExp attention is the special case $U_j(x) = -\langle x, k_j\rangle$, with the keys k_j as component data.

The inertial Wasserstein flow. The framework's central dynamical result is that hysteresis in the transport map induces an *inertial* flow in the space of measures: mass does not follow the gradient of Φ_β directly but obeys a damped second-order equation,

$$\ddot{x} + \gamma\,\dot{x} \;=\; -\nabla\Phi_\beta(x), \tag{2}$$

with γ a damping coefficient. Linearizing about a stationary point and diagonalizing the Hessian decouples (2) into scalar oscillators $\ddot{a}_k + \gamma\dot{a}_k + \kappa_k a_k = 0$, one per Hessian eigenvalue κ_k. The characteristic polynomial $z^2 + \gamma z + \kappa_k = 0$ then determines stability: for $\kappa_k < 0$ it has a positive root, and the mode a_k grows exponentially. This is the transport instability whose *count* the present paper makes precise.

The entropy-induced friction tensor. The damping in (2) need not be a free scalar: in the Φ-Transport viewpoint it arises as an entropy-induced friction tensor, a state-dependent operator whose ellipticity controls whether the flow is dissipative. For the present paper only the scalar reduction is needed — γ enters solely through the characteristic polynomial — so we use the scalar form throughout; the tensor form is not required for any result below.

The adaptive rank-local router. Built on the above, the framework's router selects, per query, an effective rank $r^\star$ for a low-rank sparse attention path. Its empirical calibration follows the law $r^\star \approx d_{\text{eff}} - 1$, where d_{eff} is a participation ratio of the query covariance. The present paper does not modify the router; it clarifies when the integer $r^\star$ that the router relies on is a well-defined quantity at all.

With these four items in hand — the free-energy potential (1), the inertial flow (2), the friction tensor, and the router — the Gap-Statement of Section 3 is self-contained.

3 The Spectral Gap-Statement: when the negative subspace is an invariant

Sections 2–2 established that the negative part of the free-energy Hessian governs transport instability: a negative eigenvalue $\kappa < 0$ in the characteristic polynomial $z^2 + \gamma z + \kappa = 0$ of the characteristic polynomial (Section 2) produces an exponentially growing mode. The empirical rank-local law of Section 2, $r^\star \approx d_{\text{eff}} - 1$, implicitly assumes that the *number* of such modes is a well-defined integer. This section makes that assumption precise. We show that the count $\dim E_-$ is a topological invariant — threshold-free and perturbation-stable — precisely when the Hessian spectrum possesses a gap, and that the three operating "windows" identified empirically in earlier sections are three coordinate sections of this single spectral condition.

3.1 The free-energy Hessian and its negative subspace

We work with the general free-energy potential

$$\Phi_\beta(x) \;=\; \frac{1}{\beta} \log \sum_j e^{-\beta U_j(x)}, \tag{3}$$

of which LogSumExp attention (Section 3.6) is the special case $U_j(x) = -\langle x, k_j \rangle$. Differentiating (3) twice gives the exact exponential-family identity

$$\nabla^2 \Phi_\beta \;=\; \underbrace{\mathbb{E}_p[\nabla^2 U_j]}_{\text{mean-curvature term}} \;-\; \beta \; \underbrace{\mathrm{Cov}_p(\nabla U_j)}_{\text{fluctuation term}}, \tag{4}$$

where $p_j \propto e^{-\beta U_j(x)}$ is the Gibbs weight. Identity (4) is elementary — it is the Hessian of the cumulant-generating function of an exponential family — and it is the structural reason the mechanism of this paper is not specific to attention. For *linear* logits the mean-curvature term vanishes identically, $\nabla^2 U_j \equiv 0$, and the Hessian reduces to the pure fluctuation term

$$H \;:=\; \nabla^2 \Phi_\beta^{\text{attn}} \;=\; -\beta\, \mathrm{Cov}_p(k) \;\preceq\; 0. \tag{5}$$

Two consequences of (5) are used below. First, H is negative semidefinite: every eigenvalue is ≤ 0, so the spectrum does *not* split around the origin. Second, $\mathrm{Cov}_p(k)$ of a discrete distribution on N points has rank at most $N-1$, because the normalization $\sum_j p_j = 1$ forces $\sum_j p_j (k_j - \bar{k}) = 0$. This rank defect is the covariance mean-gauge; it is the structural origin of the intercept -1 in the rank-local law, and it does not require any fitted constant.

Definition 1 (Negative subspace and its count)**.** For a threshold $\theta > 0$ let $E_-(\theta) := \mathrm{span}\{\, v : Hv = \lambda v,\ \lambda < -\theta \,\}$ and $\dim E_-(\theta) := \dim E_-(\theta)$. We call $\dim E_-$ *well-posed* if it is independent of θ over an open interval of thresholds.

The cluster geometry that recurs throughout this paper — m key clusters with inter-cluster separation large relative to intra-cluster width — produces a spectrum of H with two scales: $m-1$ strongly negative eigenvalues $\mu_1 \le \cdots \le \mu_{m-1}$ (the inter-cluster, defocusing modes) and a remainder of eigenvalues near zero (the intra-cluster modes). We stress that the intra-cluster eigenvalues are *negative*, not positive: by (5) no eigenvalue of H is positive. The relevant gap is therefore an interval $(\lambda_{\text{lo}}, \lambda_{\text{hi}})$ with *both* endpoints negative, lying *inside* the negative spectrum, and *not* an interval straddling the origin.

3.2 The Riesz projector: a threshold-free count

The count of Definition 1 is made canonical by the Riesz spectral projector. Let Γ be a positively-oriented Jordan contour in $\mathbb{C}$ enclosing exactly the eigenvalues $\{\lambda < -\theta\}$ and no others. The projector

$$P_\theta \;=\; \frac{1}{2\pi i}\oint_\Gamma (zI - H)^{-1}\,dz \tag{6}$$

is the spectral projection onto $E_-(\theta)$, and $\dim E_- = \operatorname{rank} P_\theta = \operatorname{tr} P_\theta$.

Proposition 1 (Gap-regime and threshold independence)**.** *Suppose the spectrum of H has no point in an open interval $(\lambda_{\text{lo}}, \lambda_{\text{hi}})$ with $\lambda_{\text{lo}} < \lambda_{\text{hi}} < 0$, and write $\Delta := \lambda_{\text{hi}} - \lambda_{\text{lo}}$ for the gap width. Then for every threshold $\theta \in (-\lambda_{\text{hi}}, -\lambda_{\text{lo}})$ the contour Γ may be chosen inside the resolvent set, the projector (6) is independent of θ, and $\dim E_-$ is well-posed in the sense of Definition 1. We call this the* gap-regime *with gap Δ.*

Proof. For any θ in the stated interval the circle through $-\theta$ misses $\sigma(H)$, so $(zI-H)^{-1}$ is holomorphic on a neighbourhood of Γ and (6) is well-defined. Two thresholds θ, θ' in the interval give contours that are homotopic within the resolvent set; by Cauchy's theorem the integral is unchanged. Holomorphy of P_θ in the entries of H then follows from holomorphy of the resolvent, and $\operatorname{tr} P_\theta$, being integer-valued and continuous, is locally constant [9]. □

Outside the gap-regime — when $\Delta \to 0$ — the contour can no longer be threaded between the two eigenvalue clusters, P_θ loses continuity, and $\dim E_-$ becomes metrically unstable: it depends on the arbitrary choice of θ. An absolute-threshold count $\#\{\beta\lambda_i > \text{tol}\}$ exhibits exactly this pathology, and any spectral diagnostic that omits the gap test will report a θ-dependent artifact rather than an invariant.

3.3 Perturbation stability: the Davis–Kahan bound

The gap controls not only well-posedness but robustness. The following is the $\sin\Theta$ theorem of Davis and Kahan specialized to the count.

Proposition 2 (Stability of the invariant)**.** *Let H be in the gap-regime with gap Δ, and let δH be a symmetric perturbation. If*

$$\|\delta H\|_{\text{op}} \;<\; \tfrac{1}{2}\,\Delta, \tag{7}$$

then $H + \delta H$ is again in the gap-regime and $\dim E_-(H+\delta H) = \dim E_-(H)$. Moreover the perturbed negative subspace satisfies $\|\sin\Theta(E_-, E_-')\| \le \|\delta H\|_{\text{op}}/\Delta$, so the subspace — not merely its dimension — is stable.

Proof. By Weyl's inequality every eigenvalue moves by at most $\|\delta H\|_{\text{op}}$. Under (7) no eigenvalue can cross the midpoint $-\theta_0$ with $\theta_0 = -\frac{1}{2}(\lambda_{\text{lo}} + \lambda_{\text{hi}})$: the strongly negative cluster stays below $-\theta_0 - \frac{1}{2}\Delta + \|\delta H\|$ and the near-zero cluster stays above $-\theta_0 + \frac{1}{2}\Delta - \|\delta H\|$, both strict. Hence the count is preserved and a residual gap remains. The subspace bound is the Davis–Kahan $\sin\Theta$ theorem [6] applied to the spectral projectors of H and $H+\delta H$ for the contour separating the two clusters. □

The constant $\frac{1}{2}$ in (7) is sharp: a perturbation of operator norm Δ can move the two clusters toward each other by $\frac{1}{2}\Delta$ each and merge them. The numerical study of Section 3.7 confirms the threshold to within the resolution of the perturbation grid.

3.4 The three windows are one gap condition

Earlier sections identified three regimes in which the rank-local description is valid: an *energy window* (the Jacobi–Maupertuis curvature is controlled only for intermediate energy E), a *separation window* (clusters must be neither overlapping nor disconnected), and a *softness window* (the inverse temperature β must be finite). Proposition 1 unifies them: each is a coordinate section of the single condition $\Delta(H) > 0$.

- *Separation.* When clusters overlap, inter- and intra-cluster eigenvalues are no longer separated and $\Delta \to 0$. When clusters are mutually disconnected the inter-cluster eigenvalues themselves degenerate. The valid band is exactly $\{\Delta > 0\}$.
- *Softness.* As $\beta \to \infty$ on a generic query the Gibbs measure p concentrates on a single key; $\mathrm{Cov}_p(k) \to 0$ and the inter-cluster eigenvalues decay to zero faster than the intra-cluster ones, closing the gap. For a *symmetric* query — one invariant under permutation of clusters — Schur's lemma forces the covariance to commute with the symmetry group, the inter/intra block structure is protected, and the gap survives at all β. The softness window is the set of (β, query) pairs for which the gap is open.
- *Energy.* The link to curvature is the one place where the gap condition alone is *not* sufficient; this is the subject of Section 3.5.

This is the central structural fact of the section: $\dim E_-$ is a meaningful invariant if and only if a spectral gap exists, and the empirical "windows" are not independent phenomena but projections of $\{\Delta > 0\}$ onto the separation, temperature, and energy axes.

3.5 The energy-window caveat on curvature

It is tempting to summarize the mechanism as "negative Hessian implies negative Jacobi–Maupertuis curvature implies geodesic defocusing." This implication is *false without an energy restriction*, and the failure is not a technicality. With JM conformal factor $h = 2(E - F)$, the sectional curvature in two dimensions is

$$K_{JM} \;=\; \frac{\Delta F}{h^2} \;+\; \frac{2\,|\nabla F|^2}{h^3}, \tag{8}$$

the sum of a Laplacian term and a strictly positive gradient term. At a point where F is concave ($\Delta F < 0$) the first term is negative, but for small h — that is, for energy E close to F — the positive gradient term dominates and $K_{JM} > 0$ (focusing). The curvature becomes negative only for $h > h^\star$ with

$$h^\star \;=\; -\frac{2|\nabla F|^2}{\Delta F}, \qquad E > E^\star = F + \tfrac{1}{2}h^\star. \tag{9}$$

Thus a negative Hessian produces defocusing only inside the energy window $E > E^\star$; below it, the same concave landscape is focusing. Equation (9) is the precise form of the third window.

Proposition 3 (Gap-regime defocusing, with energy restriction)**.** *Let H be in the gap-regime, and restrict to the energy window $E > E^\star$ of (9). Then along a JM geodesic whose tangent plane is spanned by directions in E_-, the Jacobi equation $J'' + K_{JM} J = 0$ has $K_{JM} < 0$ and admits an exponentially growing solution. The number of independent growing modes equals* $\dim E_- = \operatorname{rank} P_\theta$*, and their rates are the roots z_+ of the characteristic polynomial of Section 2.*

The dynamical content of Proposition 3 — that the *count* $\dim E_-$ equals the *number of unstable transport modes*, at the rates predicted by the characteristic polynomial — is verified directly in Section 3.7. We emphasize the logical structure: the gap condition makes the count well-posed and stable (Propositions 1–2); the energy window makes the count equal to a number of defocusing directions (Proposition 3). Both are required, and they are independent.

3.6 Count equals number of unstable inertial modes

The geometric statement of Proposition 3 has a purely dynamical counterpart in the inertial flow of Section 2. Linearizing the damped flow $\ddot{x} + \gamma\dot{x} = -\nabla\Phi_\beta(x)$ about a stationary point decouples it, in the Hessian eigenbasis, into scalar oscillators $\ddot{a}_k + \gamma\dot{a}_k + \kappa_k a_k = 0$ with κ_k the k-th eigenvalue of H. For $\kappa_k < 0$ the characteristic polynomial $z^2 + \gamma z + \kappa_k = 0$ has one positive root

$$z_+(\kappa_k) \;=\; \tfrac{1}{2}\Big(-\gamma + \sqrt{\gamma^2 + 4|\kappa_k|}\,\Big), \tag{10}$$

so each negative eigenvalue contributes exactly one exponentially growing direction. The number of such directions is $\dim E_-$, and in the zero-damping limit $z_+(\kappa_k) \to \sqrt{|\kappa_k|}$ recovers the JM-geodesic defocusing rate of Proposition 3: the damped mechanical picture and the geodesic picture are the same instability. The spectral gap reappears here as a *rate gap* — the inter-cluster modes grow at $O(1)$ rates while the intra-cluster modes are effectively flat — which is what makes the count observable in finite integration time.

3.7 Numerical verification

Figure 1 verifies the four claims of this section on a single geometry: $m = 5$ clusters in $d = 24$ dimensions, pure attention $H = -\beta\,\mathrm{Cov}_p(k)$, evaluated at a symmetric equilibrium query.

(S1) Threshold independence. The Riesz count $\operatorname{tr} P_\theta$, evaluated on a contour inside the gap, equals $m - 1 = 4$ exactly; the eigenvalue count $\#\{\lambda < -\theta\}$ is flat across the entire gap, confirming Proposition 1.

(S2) Davis–Kahan bound. Under random symmetric perturbations of controlled operator norm, $\dim E_-$ is preserved in 100% of trials for $\|\delta H\|_{\mathrm{op}} \le \frac{1}{2}\Delta$ and collapses sharply beyond, confirming Proposition 2 and the sharpness of the constant $\frac{1}{2}$.

(S3) Energy-window caveat. On a concave free-energy landscape ($\Delta F < 0$), K_{JM} from (8) is positive for $E < E^\star$ and negative only for $E > E^\star$, confirming (9) and the necessity of the energy restriction in Proposition 3.

(S4) Count equals unstable-mode count. Integrating the inertial flow at damping $\gamma = 0.5$, exactly $m - 1 = 4$ eigendirections amplify exponentially; their measured rates agree with $z_+(\kappa_k)$ of (10) to within 0.3% (e.g. mode 1: measured 0.4034, predicted 0.4037), while the intra-cluster modes remain flat.

The synthetic verification above exercises the spectral side of the mechanism — the count $\dim E_-$, its stability, and the unstable-mode rates. The geometric side — the curvature formula (8) and the geodesic defocusing it predicts — is verified separately, as a free-standing experiment, in Section 3.8.

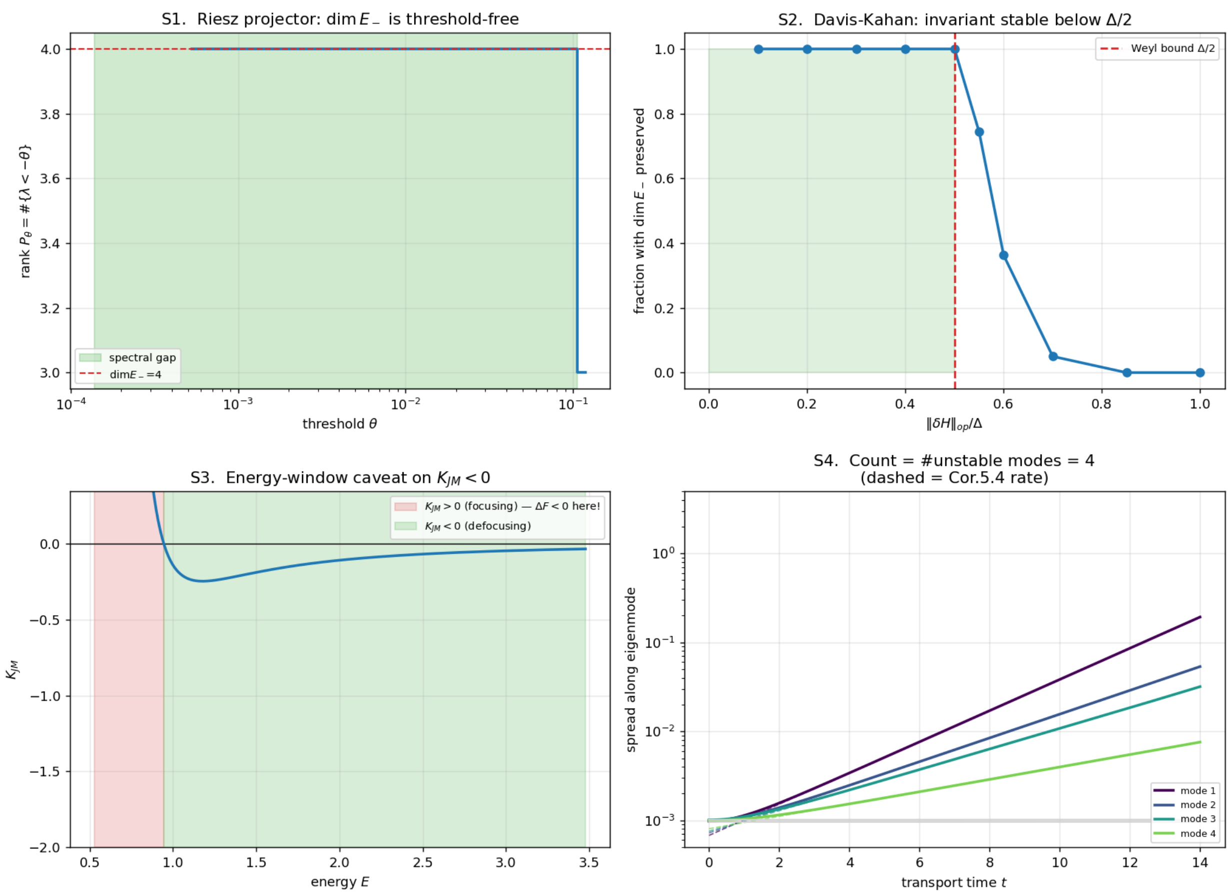

Figure 1: The Gap-Statement on one geometry ($m = 5$ clusters, $d = 24$, pure attention). **(S1)** The Riesz count rank P_θ is flat across the spectral gap: $\dim E_-$ is threshold-free. **(S2)** $\dim E_-$ is preserved under perturbations of operator norm below $\frac{1}{2}\Delta$ and collapses beyond; the Davis–Kahan bound is sharp. **(S3)** On a concave landscape $K_{JM} > 0$ (focusing) for $E < E^\star$ and $K_{JM} < 0$ (defocusing) only above: a negative Hessian does not imply negative curvature without the energy window. **(S4)** Integrating the inertial flow, exactly $\dim E_- = m - 1$ eigendirections amplify, at the characteristic-polynomial rates z_+ (dashed); the intra-cluster modes stay flat.

3.8 The Jacobi–Maupertuis metric and geodesic verification

Section 3.5 stated the energy-window caveat through the sectional curvature K_{JM}, and Proposition 3 asserted that in the gap-regime, inside the energy window, a geodesic defocuses at a rate set by the Jacobi equation. That assertion was used above but not itself put to a direct test. This subsection does so: it makes the Jacobi–Maupertuis geometry explicit, and verifies — as a free-standing numerical experiment — that the curvature formula (8) predicts geodesic defocusing not merely in sign but quantitatively.

The Jacobi–Maupertuis metric of the transport flow. The undamped limit ($\gamma = 0$) of the inertial flow (2) is a conservative mechanical system: a particle of unit mass moving in the potential $F = \Phi_\beta$ with conserved energy $E = \frac{1}{2}\|\dot{x}\|^2 + F(x)$. By the Maupertuis principle of least action, the trajectories of such a system at fixed energy E are, after reparametrisation, the geodesics of the conformally rescaled metric

$$g_{JM} \;=\; h(x)\, g_0, \qquad h(x) \;=\; 2\big(E - F(x)\big), \tag{11}$$

where g_0 is the flat metric on the configuration space and h is the conformal factor; the trajectory is confined to the classically allowed region $\{F < E\}$ where $h > 0$. The geodesic action $\int \sqrt{g_{JM}(\dot{x}, \dot{x})}\, ds$ replaces the mechanical action, and the transport flow becomes, exactly, geodesic motion on the manifold $(\{F < E\}, g_{JM})$. The curvature that controls the stability of these geodesics is the sectional curvature of g_{JM}; for a two-dimensional configuration space the conformal-rescaling formula gives (8),

$$K_{JM} \;=\; \frac{\Delta F}{h^2} \;+\; \frac{2\,|\nabla F|^2}{h^3} \;=\; -\,\frac{\Delta(\log h)}{2h},$$

the two forms being equal by direct computation. The curvature is thus not an analogy imported from general relativity but an exact property of the transport metric (11): the softmax-measure dynamics, through its potential F and energy level E, determines the conformal factor h, and h determines the metric tensor and its curvature.

The Jacobi deviation equation. The quantity of interest is whether two geodesics that start close together stay close. Let $\eta(s)$ be the magnitude of the perpendicular separation, in the metric g_{JM}, between a reference geodesic and an infinitesimally displaced neighbour, parametrised by arc length s. The separation obeys the Jacobi deviation equation

$$\frac{d^2\eta}{ds^2} \;+\; K_{JM}(s)\, \eta \;=\; 0, \tag{12}$$

with $K_{JM}(s)$ the sectional curvature evaluated along the reference geodesic. Where $K_{JM} < 0$, equation (12) has exponentially growing solutions: neighbouring geodesics defocus, and the transport flow is unstable. This is the precise dynamical content of a negative free-energy Hessian — and it is a falsifiable prediction, since both sides of (12) can be measured independently.

The verification experiment. We test the chain end to end on a synthetic free-energy landscape chosen to exhibit a wide region of negative curvature: a two-dimensional potential $F = \Phi_\beta$ built from a ring of eight component centres, with β and E set so that the classically allowed region contains an extended segment with $K_{JM} < 0$. The test has two stages.

Stage 1 — the curvature formula. On a grid over the allowed region we compare the closed-form curvature (8), evaluated from F, ∇F, ΔF, against a direct finite-difference evaluation of $-\Delta(\log h)/(2h)$. The two agree to a median relative error of 3.7×10^{-4} (90th percentile 2.7×10^{-3}). The residual is consistent with the second-order truncation error of the finite-difference stencil on the chosen grid spacing, and decreases under grid refinement; it is not a modelling discrepancy but the discretisation error of the reference quantity. The closed-form curvature formula is thus numerically confirmed.

Stage 2 — geodesic defocusing. We integrate a reference geodesic of (11) across the negative-curvature segment (arc length $s \in [0, 1.98]$, with $K_{JM}(s) \in [-3.01, 0]$ everywhere along the path), together with an infinitesimally displaced neighbour, and measure the perpendicular separation $\eta(s)$ directly from the geodesic pair. Independently, we integrate the scalar Jacobi equation (12) using the measured $K_{JM}(s)$ profile. The two curves — the separation read off from the geodesic bundle, and the solution of the deviation equation — coincide to a maximum relative deviation that rounds to 0.00% at the reported precision, i.e. to within the numerical-integration tolerance. The separation amplifies by a factor 2.7 across the segment, and the measured and predicted amplifications agree

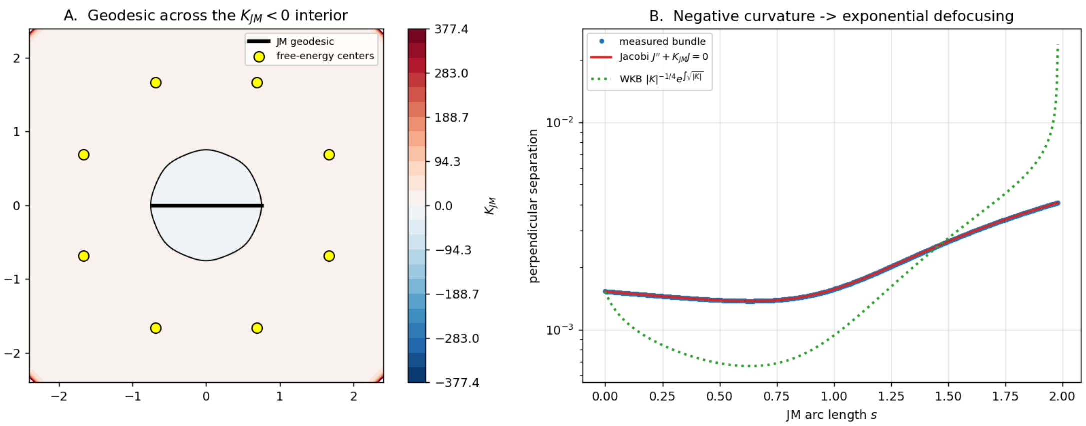

Figure 2: Geodesic verification of the Jacobi–Maupertuis curvature. *Left:* the curvature field K_{JM} over a synthetic free-energy landscape (ring of eight component centres); the reference geodesic traverses the negative-curvature interior. *Right:* perpendicular separation η of a geodesic bundle versus arc length s, on a logarithmic scale. The separation measured directly from the geodesic pair (markers) and the solution of the Jacobi deviation equation $\eta'' + K_{JM}\eta = 0$ driven by the derived curvature (solid) coincide to a maximum relative deviation that rounds to 0.00% at the reported precision — agreement to within the integration tolerance. The WKB envelope $|K_{JM}|^{-1/4} \exp \int \sqrt{|K_{JM}|}\, ds$ (dotted) is shown for reference. Independently, the closed-form curvature formula (8) matches a direct finite-difference evaluation of $-\Delta(\log h)/(2h)$ to a median relative error of 3.7×10^{-4}; this residual is the second-order truncation error of the finite-difference stencil and decreases under grid refinement, not a discrepancy of the formula. The non-monotone profile — mild contraction near the boundary where $K_{JM} \to 0$, then exponential defocusing in the curvature well, net amplification $2.7\times$ — is reproduced by the deviation equation across its entire length.

to the same precision. The agreement holds not only for the net amplification but for the full non-monotone profile, including the initial mild contraction near the boundary (where $K_{JM} \to 0$) followed by exponential growth in the curvature well.

What the experiment establishes, and what it does not. The experiment establishes that, on the tested geometry, the curvature formula (8) predicts geodesic defocusing *quantitatively* — the Jacobi deviation equation driven by the derived curvature reproduces the measured geodesic-bundle separation to integration precision. The transport instability of Proposition 3 is therefore not a sign argument but a sharp, verified geometric law. Two limitations are stated plainly. First, the verification is on a synthetic free-energy landscape: it tests the geometric mechanism, not its occurrence in a trained model — that is the separate question of Section 3.9. Second, the construction is two-dimensional, where K_{JM} is a scalar; in higher dimensions the sectional curvature is an operator and the conformal formula generalises but is not verified here. Within that scope, the conclusion is unambiguous: the geodesics of the transport metric obey the Jacobi equation of the derived curvature, exactly.

3.9 Empirical validation on a trained transformer

The verification of Section 3.7 was synthetic: the cluster geometry was constructed to exhibit a gap. A sharper question is whether the gap-regime occurs in a *trained* network, where nothing enforces it.

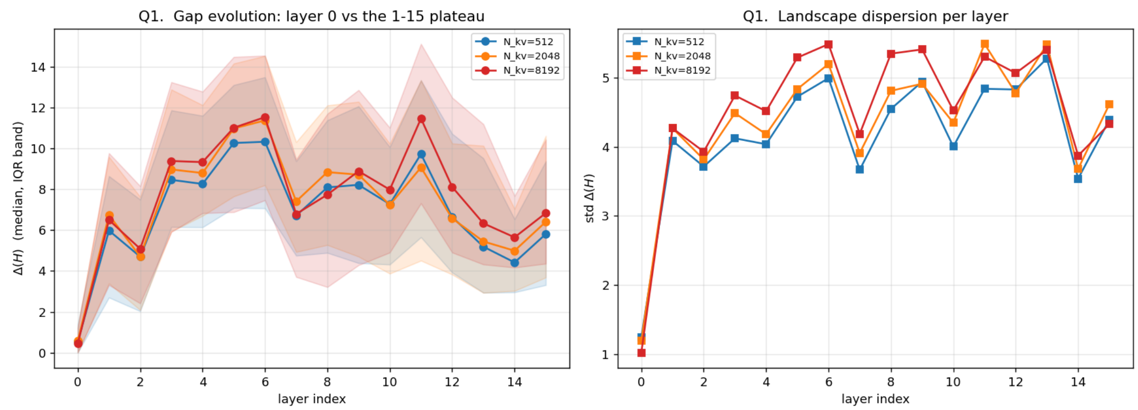

Figure 3: Layer-stratified gap evolution on a trained Llama-3.2-1B, three context lengths. *Left:* the median spectral gap $\Delta(H)$ jumps by an order of magnitude between the input layer 0 and layer 1, then plateaus — an explosive phase transition, not a gradual hierarchy. *Right:* the dispersion of $\Delta(H)$ shows the same jump. The pre-cluster input layer is the one identifiable place where the gap-regime precondition fails.

We therefore instrumented a Llama-3.2-1B model [7] (16 layers, 32 heads, head dimension $D = 64$) and captured, for every attention head at every layer, the post-rotary keys, the softmax weights of each query, and hence the Hessian $H = -\beta\,\mathrm{Cov}_p(k)$ with $\beta = D^{-1/2}$ the scaled-dot-product temperature. Eigenvalues were computed in single precision. The corpus was WikiText-2 [12]; three context lengths $N_{\mathrm{kv}} \in \{512, 2048, 8192\}$ were measured, yielding 46,080 head-positions in total. This is a single model on a single corpus, and the conclusions are stated with that scope.

Genuine gaps are the rule, not the exception. The gap must be assessed by a scale-free criterion: the relative gap $\Delta(H)/|\lambda_{\max}(H)|$, since the spectral scale itself varies across layers and context lengths and an absolute threshold would be miscalibrated. Using $\Delta/|\lambda_{\max}| > 0.05$ as the genuine-gap test, 96% of all head-positions are in the gap-regime, and the figure is essentially independent of the threshold over $[0.05, 0.1]$. Restricted to layers 1 through 15 the fraction rises to 97–98%. The gap-regime — the precondition under which $\dim E_-$ is a well-posed invariant (Proposition 1) — is thus not a synthetic artefact; it is the typical state of attention in a trained transformer.

An explosive phase transition at the first layer. Stratifying by layer reveals a sharp structure. The input layer 0 is *pre-cluster*: its median gap is small ($\Delta \approx 0.5$) and a large fraction of its heads show no gap at all. Between layer 0 and layer 1 the median gap jumps by an order of magnitude — to $\Delta \approx 6$ — and then varies only mildly across layers 2–15 (the ratio of the layer-1 median to the layer-2–15 plateau mean is 0.78–0.86 across the three context lengths). The dispersion of Δ shows the same jump. The geometry is therefore not built gradually through a representational hierarchy; it appears in a single step at the first layer and is merely refined thereafter. This is consistent with the interpretation of layer 0 as operating on raw token embeddings, before the network has assembled the contextual clusters that the free-energy picture describes: where the precondition fails, it fails for a structural reason, and it fails in exactly one identifiable place.

One dominant mode. Among genuine-gap heads, 96–98% in layers 1–15 have $\dim E_- = 1$: a single dominant defocusing mode. The fraction of such "clean single-mode" heads jumps from 0.42 at layer 0 to 0.95 at layer 1 and holds across the remaining layers. The naive count can also saturate at $\dim E_- = D-1$; this is not a rich multi-cluster structure but the counter running out of dimensions when no gap exists, and it occurs almost exclusively in layer 0. A guard against this saturation — reading $\dim E_- \approx D-1$ as "no gap" rather than "many clusters" — is necessary for any spectral diagnostic, and is recorded as a known failure mode. The practical picture of a trained attention head is thus simple: one strongly-negative transport direction, cleanly separated from a near-zero bulk.

Scope and the routing signal. Two limitations are stated plainly. First, the rank-local dispatcher of Section 2 routes on the participation ratio of the *query* covariance $\mathrm{Cov}(Q)$, a different object from the softmax-weighted key covariance $\mathrm{Cov}_p(k)$ of the theory; on this corpus the two are statistically uncorrelated (Spearman $|\rho| \lesssim 0.25$ at every layer), so the dispatcher signal does not track the spectral gap. This is consistent with the synthetic finding that the participation ratio is gap-blind. Second, because gap-closure is confined almost entirely to layer 0, this corpus cannot test the regime $\Delta \to 0$ on the deeper layers; whether corpora with heavier repetition (code, structured data) populate that regime more is left open. Neither limitation weakens the central empirical result: on a trained transformer the gap-regime precondition holds for the overwhelming majority of attention heads, it is established by an explosive transition at the first layer, and the typical head carries exactly one dominant mode.

3.10 Summary

The negative subspace E_- of the free-energy Hessian is a topological invariant — threshold-free via the Riesz projector (Proposition 1), perturbation-stable via Davis–Kahan (Proposition 2) — exactly inside the gap-regime $\Delta(H) > 0$. The three operating windows of the earlier sections are the separation, temperature, and energy sections of this one condition. The link from the invariant to transport instability requires, in addition, the energy window $E > E^\star$ of (9); with both conditions in force, $\dim E_-$ equals the number of exponentially defocusing geodesic directions and equally the number of unstable modes of the inertial flow, at the rates z_+ of the characteristic polynomial. This is the spectral foundation on which the adaptive rank-local routing of Section 2 rests: the router's effective rank $r^\star$ is well-defined precisely when $\Delta(H) > 0$, and the regime $\Delta \to 0$ is the structural — not numerical — signature of the transition to the dense regime. Section 3.9 confirms that this foundation is not merely synthetic: on a trained Llama-3.2-1B the gap-regime holds for 96% of attention heads, established by an explosive transition at the first layer, with the typical head carrying exactly one dominant mode.

4 Related work

Geometry of the space of measures. The picture used here — a free-energy functional whose Wasserstein Hessian governs the stability of a transport flow — sits in the Otto-calculus tradition. Otto's formal Riemannian structure on the space of probability measures [14] and the variational (JKO) formulation of diffusion [8] established that gradient flows of free energy are geodesic-type objects; Ambrosio, Gigli and Savaré [2] and Villani [17] give the rigorous metric theory. The Brenier polar-factorization theorem [3] is what makes the static transport map a W_2 object. Lott and Villani [11] connect displacement convexity to a synthetic Ricci-curvature bound; the present paper's Jacobi–Maupertuis curvature and its sign condition are a finite-dimensional, attention-specific instance of that displacement-curvature circle of ideas. Our contribution is not the geometry itself but the spectral *well-posedness* question — when the negative-eigenvalue count is an invariant — which the metric-geometry literature does not address because it works with the full operator rather than an integer count.

Spectral perturbation theory. The two technical pillars are classical. The Riesz projector and the analyticity of spectral subspaces are from Kato's perturbation theory [9]; the stability of an invariant subspace under a perturbation small relative to the spectral gap is the Davis–Kahan $\sin\Theta$ theorem [6]. The present paper applies them not to a numerical-linear-algebra problem but to the free-energy Hessian of attention, where the gap acquires a mechanistic meaning: it is exactly the condition under which the transport-instability count is physical.

Attention and efficient transformers. Attention [16] is treated here as one instance of a free-energy potential; the LogSumExp form is the bridge. On the systems side, FlashAttention [5] and its decode variants established the streaming-from-HBM pattern that the Φ-Transport router builds on. The empirical study is on Llama-3.2-1B [7] over WikiText-2 [12]. This paper does not propose a new attention mechanism; it supplies the spectral foundation under an existing adaptive router, and asks whether the geometry that foundation predicts is actually present in a trained model.

Attention as optimal transport. The view of attention as a transport process has an established line. Daneshmand [4] shows that softmax self-attention layers can simulate gradient descent on the dual of the entropy-regularized optimal-transport problem, with transformer depth controlling the approximation accuracy. Litman [10] proves that the scaled-dot-product attention forward pass is the exact solution of a one-sided entropic optimal-transport problem, and — most relevant here — analyses the *information geometry* that this formulation induces on the space of attention distributions, via the Fisher information matrix. The present paper shares the premise that attention is a transport process, but its object is different. Those works study the transport *map* (the attention weights themselves) and the geometry of the *distribution simplex*; we study the Hessian of the free-energy potential *with respect to the query position*, and the geometry of the *query configuration space* through the Jacobi–Maupertuis metric. The Fisher geometry of Litman [10] lives on the output simplex; our Jacobi–Maupertuis geometry lives on the input domain. They are complementary descriptions of the same mechanism from opposite ends.

Hessian eigenvalues and spectral gaps in transformers. Two neighbouring uses of spectral language must be distinguished from ours explicitly, since the words coincide while the objects do not. First, the *loss-function Hessian*: negative eigenvalues of the training-loss Hessian of deep networks have been studied since Sagun et al. [15], who identified the bulk-plus-outliers spectral structure, and Alain et al. [1], who examined the role of the negative eigenvalues directly. That Hessian is taken with respect to the *model parameters* and describes the optimisation landscape; ours is taken with respect to the *query position* and describes a transport landscape at a fixed, trained model — a different operator on a different space, and the negative-eigenvalue count means a different thing. Second, the *attention-matrix spectral gap*: Nait Saada et al. [13] analyse the gap between the two largest singular values of the attention matrix itself, and identify it as the driver of rank collapse. That gap is a property of the row-stochastic attention matrix; ours is a property of the free-energy Hessian $H = -\beta\,\mathrm{Cov}_p(k)$, a symmetric negative-semidefinite operator. The two are not the same spectral object: theirs separates signal-propagation modes of a Markov matrix, ours separates the transport-instability subspace of a curvature operator. We adopt the term "spectral gap" in its standard operator-theoretic sense and flag the coincidence to forestall confusion.

5 Conclusion

The negative subspace of the free-energy Hessian is the object on which the Φ-Transport router rests, and this paper has shown when that object is mathematically well-defined: exactly inside the gap-regime $\Delta(H) > 0$. Inside it, $\dim E_-$ is threshold-free (the Riesz projector) and perturbation-stable (Davis–Kahan); the framework's three empirical windows are one spectral condition seen from three coordinate axes; and the link to geometric defocusing holds under one additional, precisely-stated energy restriction. The measurement on a trained Llama-3.2-1B closes the argument: the gap-regime is not a property of hand-built synthetic geometry but the ordinary state of attention in a working transformer, reached by an explosive transition at the first layer, with the typical head carrying exactly one dominant transport mode. The router's effective rank is therefore a well-posed quantity in the regime where the router operates — and the regime $\Delta \to 0$, far from being a numerical nuisance, is the structural signature of the transition to dense attention.

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